

The empirical recurrence for OEIS A234862, recovered from its tail

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Abstract

OEIS A234862 records “Empirical recurrence of order 70” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 100$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{70} = -141557760 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A234862 is “Number of $(n+2) \times (1+2) \dots 3$ arrays with each 3×3 subblock having the sum of its 72 absolute element differences equal to 52, and no adjacent elements equal.”. It has offset 1 and begins

2216, 11714, 61954, 326786, 1791552, 9897540, 54119058, 297874488, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:28:41, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 100$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 100$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 200$ exact terms from $n = 2$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	1	c_{19}	−3978810	c_{37}	−115607711808	c_{55}	533550171136
c_2	0	c_{20}	−23968772	c_{38}	361718025536	c_{56}	−449839740928
c_3	207	c_{21}	−140064414	c_{39}	985314541360	c_{57}	−325029820416
c_4	−147	c_{22}	179766944	c_{40}	304679059232	c_{58}	22473136128
c_5	58	c_{23}	214835352	c_{41}	−579734990272	c_{59}	353544454144
c_6	−9096	c_{24}	1182874974	c_{42}	−1850252798656	c_{60}	−117915000832
c_7	5669	c_{25}	−546706820	c_{43}	−570794335488	c_{61}	−272490676224
c_8	−5974	c_{26}	−852001680	c_{44}	804027425856	c_{62}	172891471872
c_9	162730	c_{27}	−5030300380	c_{45}	2112960527936	c_{63}	30019239936
c_{10}	−105215	c_{28}	−890287888	c_{46}	544864498688	c_{64}	−75358715904
c_{11}	103014	c_{29}	−609988512	c_{47}	−1077037879168	c_{65}	2514747392
c_{12}	−1437127	c_{30}	4044109836	c_{48}	−1601375576064	c_{66}	−4030726144
c_{13}	1174748	c_{31}	3348929392	c_{49}	120131140864	c_{67}	−6815219712
c_{14}	−532194	c_{32}	26099543800	c_{50}	927980663040	c_{68}	1396441088
c_{15}	6359301	c_{33}	76375781576	c_{51}	585414868736	c_{69}	932970496
c_{16}	−6211920	c_{34}	21511386720	c_{52}	−200025127936	c_{70}	−141557760
c_{17}	1356480	c_{35}	−125098518048	c_{53}	−657545035264		
c_{18}	−5010942	c_{36}	−399279098032	c_{54}	434542153728		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A234862 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{70-1} - \dots - c_{70}$. Its constant term is $-c_{70} = 141557760$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with

$\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -141557760 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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